# MA1522 Linear Algebra for Computing

## Lesson 16: Eigenvalues and Eigenvectors

# Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson16` (e.g. `A1234567Z_lesson16.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 16**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

# Pollev



# Section 6.1 Eigenvalues and Eigenvectors

Let $A$ be a square matrix of order $n$.

### Definition
A real number $\lambda$ is an **eigenvalue** of $A$ if there is a nonzero vector $\mathbf{v} \in \mathbb{R}^n$ such that $A\mathbf{v} = \lambda\mathbf{v}$.
Such a $\mathbf{v}$ is an **eigenvector** of $A$ associated to $\lambda$.
The **characteristic polynomial** of $A$ is the degree $n$ polynomial $\det(xI - A)$.

### Theorem
$\lambda$ is an eigenvalue of $A$ if and only if $\lambda$ is a root of the characteristic polynomial $\det(xI - A)$.

### Theorem
$A$ is invertible if and only if 0 is not an eigenvalue of $A$.

# Section 6.1 Eigenspaces and Multiplicities

Let $\lambda$ be an eigenvalue of a square matrix $A$ of order $n$.

## Definition

The **algebraic multiplicity** $r_\lambda$ of $\lambda$ is the largest integer such that

$$\det(xI - A) = (x - \lambda)^{r_\lambda} p(x)$$

for some polynomial $p(x)$.

## Definition

The **eigenspace** of $A$ associated to $\lambda$ is

$$E_\lambda = \{\, \mathbf{v} \in \mathbb{R}^n \mid A\mathbf{v} = \lambda\mathbf{v} \,\} = \text{Null}(\lambda I - A).$$

The **geometric multiplicity** of $\lambda$ is $\dim(E_\lambda) = \text{nullity}(\lambda I - A)$.

# Question 1

Let $J$ be the Jordan canonical form with three blocks: a $2 \times 2$ block with eigenvalue 2, a $1 \times 1$ block with eigenvalue 2, and a $1 \times 1$ block with eigenvalue 3,

$$J = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

1. Find the characteristic polynomial of $J$.

2. Find the geometric multiplicities of the eigenvalues 2 and 3.

# Question 1 — solution

1. $xI - J$ is upper triangular, so its determinant is the product of the diagonal entries:

$$\det(xI - J) = (x-2)(x-2)(x-2)(x-3) = (x-2)^3(x-3).$$

2. The geometric multiplicity of $\lambda$ is $\text{nullity}(\lambda I - J) = 4 - \text{rank}(\lambda I - J)$:

$$2I - J = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \text{ rank } 2, \qquad 3I - J = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \text{ rank } 3.$$

So $\dim(E_2) = 2$, with $E_2 = \text{span}\{\mathbf{e}_1, \mathbf{e}_3\}$, and $\dim(E_3) = 1$, with $E_3 = \text{span}\{\mathbf{e}_4\}$. □

# Question 2

Let

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 3 & 0 & -3 \\ -1 & 1 & 4 \end{pmatrix}.$$

1. Find the characteristic polynomial of $A$, and factorise it.

2. Find the eigenvalues of $A$ and their algebraic multiplicities. Is $A$ invertible?

3. For each eigenvalue $\lambda$ of $A$, find a basis for the eigenspace $E_\lambda$, and the geometric multiplicity of $\lambda$.

# Question 2 — solution: part 1

1. Expand $\det(xI - A)$ along the first row:

$$\begin{vmatrix} x-1 & -2 & -2 \\ -3 & x & 3 \\ 1 & -1 & x-4 \end{vmatrix} = (x-1)\begin{vmatrix} x & 3 \\ -1 & x-4 \end{vmatrix} + 2\begin{vmatrix} -3 & 3 \\ 1 & x-4 \end{vmatrix} - 2\begin{vmatrix} -3 & x \\ 1 & -1 \end{vmatrix}$$

$$\begin{aligned} &= (x-1)(x^2-4x+3) + 2(-3x+9) - 2(3-x) \\ &= (x-1)(x-1)(x-3) - 6(x-3) + 2(x-3) \\ &= (x-3)\left[(x-1)^2 - 4\right] = (x-3)(x-3)(x+1). \end{aligned}$$

So $\det(xI - A) = (x-3)^2(x+1)$.

# Question 2 — solution: part 2

2. Find the eigenvalues of $A$ and their algebraic multiplicities. Is $A$ invertible?

The eigenvalues are the roots of $(x-3)^2(x+1)$:

$$\lambda = 3 \text{ with } r_3 = 2, \qquad \lambda = -1 \text{ with } r_{-1} = 1.$$

0 is not an eigenvalue of $A$, so $A$ is invertible.

# Question 2 — solution: part 3, $\lambda = 3$

3. $E_3 = \text{Null}(3I - A)$. Row reduce:

$$3I - A = \begin{pmatrix} 2 & -2 & -2 \\ -3 & 3 & 3 \\ 1 & -1 & -1 \end{pmatrix} \xrightarrow{\text{rref}} \begin{pmatrix} 1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

With $x_2 = s$, $x_3 = t$, we get $x_1 = s + t$:

$$\begin{pmatrix} s+t \\ s \\ t \end{pmatrix} = s \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \qquad E_3 = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

$A(1,1,0)^T = (3,3,0)^T$ and $A(1,0,1)^T = (3,0,3)^T$. ✓ The geometric multiplicity of 3 is $\dim(E_3) = 2$.

# Question 2 — solution: part 3, $\lambda = -1$

$E_{-1} = \text{Null}(-I - A)$. Row reduce:

$$-I - A = \begin{pmatrix} -2 & -2 & -2 \\ -3 & -1 & 3 \\ 1 & -1 & -5 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 6 \\ 0 & -2 & -6 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}.$$

With $x_3 = t$: $(x_1, x_2, x_3) = t(2, -3, 1)$, so

$$E_{-1} = \text{span}\left\{ \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} \right\}, \qquad A \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \\ -1 \end{pmatrix}. \checkmark$$

The geometric multiplicity of $-1$ is $\dim(E_{-1}) = 1$. Each geometric multiplicity equals the algebraic multiplicity. □

# Homework

For each matrix below, find the characteristic polynomial and, for each eigenvalue $\lambda$, a basis for the eigenspace $E_\lambda$.

$$H_1 = \begin{pmatrix} 2 & 4 & 4 \\ 1 & -1 & -2 \\ -1 & -1 & 0 \end{pmatrix}, \qquad H_2 = \begin{pmatrix} 3 & 1 & -2 \\ 2 & 2 & 4 \\ -1 & 1 & 2 \end{pmatrix}, \qquad H_3 = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 5 & 0 \\ 1 & 2 & 3 \end{pmatrix},$$

$$H_4 = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 3 & 2 & 2 \\ 0 & 4 & -1 & 4 \\ 0 & 0 & -2 & 1 \end{pmatrix}, \qquad H_5 = \begin{pmatrix} 0 & 0 & 3 & 2 \\ 1 & 2 & 0 & -1 \\ 3 & 0 & -1 & -3 \\ -2 & 0 & 3 & 4 \end{pmatrix}.$$

All eigenvalues are integers.

# Question 3

Let $A$ be an $n \times n$ matrix. Suppose that $\lambda$ is an eigenvalue of $A$.

1. Show that $\lambda^k$ is an eigenvalue of $A^k$ for $k \geq 1$. If $A$ is invertible, show the same for $k \leq -1$.

2. We say a matrix $A$ is **nilpotent** if $A^k = 0$ for some $k \geq 1$. Show that if $A$ is nilpotent, then 0 is the only eigenvalue of $A$.

# Question 3 — solution: part 1

1. Let $\mathbf{v}$ be an eigenvector of $A$ associated to $\lambda$, so $\mathbf{v} \neq \mathbf{0}$ and $A\mathbf{v} = \lambda\mathbf{v}$. For $k \geq 1$, by induction,

$$A^k\mathbf{v} = A\big(A^{k-1}\mathbf{v}\big) = A\big(\lambda^{k-1}\mathbf{v}\big) = \lambda^{k-1}A\mathbf{v} = \lambda^k\mathbf{v}.$$

Since $\mathbf{v} \neq \mathbf{0}$, $\lambda^k$ is an eigenvalue of $A^k$.

Now let $A$ be invertible. Then 0 is not an eigenvalue of $A$, so $\lambda \neq 0$, and

$$\mathbf{v} = A^{-1}A\mathbf{v} = \lambda A^{-1}\mathbf{v} \quad \Longrightarrow \quad A^{-1}\mathbf{v} = \lambda^{-1}\mathbf{v}.$$

So $\lambda^{-1}$ is an eigenvalue of $A^{-1}$, with the same eigenvector $\mathbf{v}$. For $k = -m \leq -1$, apply the case $m \geq 1$ to $A^{-1}$:

$$A^k\mathbf{v} = \big(A^{-1}\big)^m\mathbf{v} = \big(\lambda^{-1}\big)^m\mathbf{v} = \lambda^k\mathbf{v}.$$

# Question 3 — solution: part 2

2. If $A$ is nilpotent, then 0 is the only eigenvalue of $A$.

Let $A^k = 0$ with $k \geq 1$.

*Every eigenvalue is* 0. If $\lambda$ is an eigenvalue of $A$ with eigenvector $\mathbf{v}$, then by part 1

$$\lambda^k \mathbf{v} = A^k \mathbf{v} = 0\,\mathbf{v} = \mathbf{0}.$$

Since $\mathbf{v} \neq \mathbf{0}$, $\lambda^k = 0$, so $\lambda = 0$.

0 *is an eigenvalue.* If not, $A$ is invertible, and then so is $A^k$. But $A^k = 0$ is not invertible.

Hence 0 is the only eigenvalue of $A$. □

# Challenge 1

Let $a_0, a_1, a_2, a_3$ be real numbers. Compute the characteristic polynomial of the following matrix:

$$C = \begin{pmatrix} 0 & 0 & 0 & -a_0 \\ 1 & 0 & 0 & -a_1 \\ 0 & 1 & 0 & -a_2 \\ 0 & 0 & 1 & -a_3 \end{pmatrix}.$$

# Next lesson

Next lesson (**Fri 23 Oct**, Sections 6.2 & 6.3), I will focus on

1. diagonalization: when a square matrix $A$ can be written as $A = PDP^{-1}$ with $D$ diagonal, and how to find $P$ and $D$;

2. **orthogonal diagonalization** of symmetric matrices.

Please watch the video clips beforehand.

# Homework — answers

| | $\det(xI - H_i)$ | eigenspaces |
|---|---|---|
| $H_1$ | $(x-2)(x-1)(x+2)$ | $E_2 = \text{span}\{(1,1,-1)^T\}$, $E_1 = \text{span}\{(0,1,-1)^T\}$, $E_{-2} = \text{span}\{(1,-1,0)^T\}$ |
| $H_2$ | $(x-4)^2(x+1)$ | $E_4 = \text{span}\{(1,1,0)^T, (-2,0,1)^T\}$, $E_{-1} = \text{span}\{(1,-2,1)^T\}$ |
| $H_3$ | $(x-2)^2(x-5)$ | $E_2 = \text{span}\{(1,0,-1)^T\}$, $E_5 = \text{span}\{(0,1,1)^T\}$ |
| $H_4$ | $(x-1)^2(x+1)(x-3)$ | $E_1 = \text{span}\{(1,0,0,0)^T, (0,1,0,-1)^T\}$, $E_{-1} = \text{span}\{(1,1,-1,-1)^T\}$, $E_3 = \text{span}\{(1,2,1,-1)^T\}$ |
| $H_5$ | $(x-2)^3(x+1)$ | $E_2 = \text{span}\{(0,1,0,0)^T, (1,0,0,1)^T\}$, $E_{-1} = \text{span}\{(1,0,-1,1)^T\}$ |

Expanded: $H_1$: $x^3 - x^2 - 4x + 4$; $H_2$: $x^3 - 7x^2 + 8x + 16$; $H_3$: $x^3 - 9x^2 + 24x - 20$; $H_4$: $x^4 - 4x^3 + 2x^2 + 4x - 3$; $H_5$: $x^4 - 5x^3 + 6x^2 + 4x - 8$.

# Challenge 1 — solution

Expand $\det(xI - C)$ along the first row. Only the $(1,1)$ and $(1,4)$ entries are nonzero:

$$\begin{vmatrix} x & 0 & 0 & a_0 \\ -1 & x & 0 & a_1 \\ 0 & -1 & x & a_2 \\ 0 & 0 & -1 & x + a_3 \end{vmatrix} = x \begin{vmatrix} x & 0 & a_1 \\ -1 & x & a_2 \\ 0 & -1 & x + a_3 \end{vmatrix} - a_0 \begin{vmatrix} -1 & x & 0 \\ 0 & -1 & x \\ 0 & 0 & -1 \end{vmatrix}.$$

The second determinant is triangular, equal to $(-1)^3 = -1$. The first has the same shape one size down; expanding it the same way gives

$$\begin{vmatrix} x & 0 & a_1 \\ -1 & x & a_2 \\ 0 & -1 & x + a_3 \end{vmatrix} = x \begin{vmatrix} x & a_2 \\ -1 & x + a_3 \end{vmatrix} + a_1 = x^3 + a_3 x^2 + a_2 x + a_1.$$

Hence

$$\det(xI - C) = x^4 + a_3 x^3 + a_2 x^2 + a_1 x + a_0.$$

□